ADICHUNCHANAGIRI UNIVERSITY

BG NAGARA, Nagamangala Taluk, Mandya District – 571448,
Karnataka, India

A DISSERTATION ON

On

"SmartLab Safety System: IoT Enabled Safety & Accountability in Laboratory"

Submitted in partial fulfillment of the requirements for the award of the degree

BACHELOR OF ENGINEERING
IN
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FACULTY OF ENGINEERING AND TECHNOLOGY
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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that the project report entitled **"SmartLab Safety System: IoT Enabled Safety & Accountability in Laboratory"** work is a bonafide work carried out by **NAZIF GANGAWALI (22ECE043), K S JAHNAVI (22ECE061), KARAN (22ECE063)** of **Faculty of Engineering & Technology (B.G.S Institute of Technology), B.G Nagara** in partial fulfillment of the award Bachelor of Engineering in **Electronics and Communication Engineering**. Under **Adichunchanagiri University**, B G Nagara during the year 2025-2026. It is certified that all corrections/suggestions indicated for the internal assessment have been incorporated in the report deposited in the department library. The project report has been approved as it satisfies the academic requirements in respect of the project work prescribed for the said degree.

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DECLARATION

We **NAZIF GANGAWALI (22ECE043)**, **K S JAHNAVI (22ECE061)**, **KARAN (22ECE063)**, students of 8th semester BE in Electronics and Communication Engineering**, B.G.S Institute of Technology (BGSIT)**, **B.G Nagara.** hereby declare that the project work entitled **"SmartLab Safety System: IoT Enabled Safety & Accountability in Laboratory"** submitted to the **ADICHUNCHANAGIRI UNIVERSITY,** during the academic year **2025- 2026**, is a record of an original work done by us under the guidance of **Dr. B N SHOBHA,** Dean & Principal, Faculty of Engineering & Technology (B.G.S Institute of Technology). This project work is submitted in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF ENGINEERING IN ELECTRONICS AND COMMUNICATION ENGINEERING,** we further declare that the work embodied in this project report has not been submitted to any other university or institute for the award of any degree.

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ABSTRACT

Modern laboratory and industrial environments face significant safety vulnerabilities when relying on isolated, legacy alarm systems or cloud-dependent internet-of-things (IoT) networks, which often suffer from high communication latency and a complete lack of personnel accountability. This project presents the design and implementation of a distributed, edge-computing IoT safety network that mitigates these critical flaws.

By deploying custom ESP32-based edge nodes across multiple physical zones, the system continuously aggregates multi-variable environmental telemetry, including combustible gas concentration (MQ-2), thermal baselines (DHT22), and spatial motion (PIR). Crucially, the architecture integrates 13.56 MHz RFID (RC522) tracking at the node level to enforce strict, real-time spatial accountability, directly correlating physical hazards with exposed personnel. To eliminate external network dependencies and latency bottlenecks, the system utilizes a localized Eclipse Mosquitto MQTT broker hosted on a central Raspberry Pi hub. Upon detecting hazard thresholds, this decentralized routing guarantees sub-200 millisecond response times, simultaneously updating a real-time WebSocket graphical dashboard and triggering a decoupled backend daemon to actuate 100W industrial relays for immediate physical warning mitigation. Ultimately, this project demonstrates a horizontally scalable, cost-effective framework that transforms passive environmental monitoring into an active, high-speed forensic safety network suitable for high-risk academic and enterprise laboratories.

TABLE OF CONTENTS

CONTENTS	TITTLE	PAGE NO.
	ACKNOWLEDGEMENT	
	ABSTRACT	
	TABLE OF CONTENTS	**i**
	LIST OF FIGURES	**ii**
CHAPTER 1	**INTRODUCTION**	1-2
	1.1 PROBLEM STATEMENT	3
	1.2 OBJECTIVE	4
CHAPTER 2	**LITERATURE SURVEY**	5-6
CHAPTER 3	**SYSTEM DESIGN AND ANALYSIS**	7-19
	3.1 EXISTING SYSTEM	8
	3.1.1 LIMITATIONS OF EXISTING SYSTEM	9
	3.2 PROPOSED SYSTEM	10-11
	3.2.1 ADVANTAGES OF PROPOSED SYSTEM	12
	3.3 SYSTEM ARCHITECTURE	13-15
	3.4 METHODOLOGY	16-18
	3.5 APPLICATIONS	19
CHAPTER 4	**SYSTEM REQUIREMENTS**	20-26
	4.1 HARDWARE REQUIREMENTS	20-24
	4.2 SOFTWARE REQUIREMENTS	25-26
CHAPTER 5	**RESULTS AND DISCUSSIONS**	27-29
	CONCLUSION	30
	BIBLIOGRAPHY	31

LIST OF FIGURES

FIGURE NO.	FIGURE NAME	PAGE NO.
Figure 1	Proposed System Mock-Up	11
Figure 2	System Architecture	13
Figure 3	Methodology Flow Chart	16
Figure 4	Raspberry Pi	20
Figure 5	ESP32 Development Board	21
Figure 6	MQ-2 Sensor	21
Figure 7	DHT22 Sensor	22
Figure 8	Motion Sensor (PIR)	22
Figure 9	Buzzer / Siren	23
Figure 10	LED Tower Lamp	23
Figure 11	RFID Module	24
Figure 12	CCTV Module & Sensors	24
Figure 13	Raspbian OS	25
Figure 14	Arduino IDE	25
Figure 15	MQTT Broker	26
Figure 16	Web Dashboard	26
Figure 17	Hardware Assembly and Physical Deployment of Node 1	27
Figure 18	Real-Time WebSocket Visualization Interface	28
Figure 19	Developed System by Team Aura x9	29

Chapter 1

INTRODUCTION

The evolution of enterprise and academic infrastructure has drastically increased the complexity of daily operations. Modern laboratories, whether academic, industrial, or research-oriented, handle a wide range of chemicals, electrical equipment, heat sources, sensors, and specialized instruments. While these environments are designed for innovation and discovery, they also carry inherent risks. The sheer volume of concurrent operations—ranging from complex chemical synthesis to high-voltage electronics testing—creates an environment where variables can fluctuate rapidly. Accidental gas leaks, unexpected fires, hazardous chemical exposure, equipment malfunction, and human errors can escalate into dangerous situations if not detected early.

Historically, mitigating these risks relied entirely on human oversight. Traditional laboratory safety systems rely heavily on manual monitoring, routine inspections, and the vigilance of staff. Standard operating procedures often dictate periodic physical walkthroughs and the logging of environmental parameters on clipboards or localized workstations. Although these practices are essential, they are often reactive rather than proactive, leaving room for delayed responses during critical moments. A human operator can only respond to a localized alarm once it has already been triggered, resulting in a critical loss of time during the earliest, most preventable stages of an incident.

The SmartLab Safety System is developed as a comprehensive solution to bridge this gap by integrating intelligent sensing, automation, and real-time alerts. Engineered by the AURA +999 team, the project aims to modernize laboratory safety using a network of sensors and microcontrollers that can continuously monitor environmental conditions and immediately respond to anomalies. By utilizing components such as gas sensors, fire detection modules, ultrasonic sensors, environmental sensors, and microcontrollers like Arduino or ESP-based systems, SmartLab becomes capable of identifying risks at their earliest stages. This proactive approach minimizes the possibility of severe accidents and significantly enhances the preparedness of lab personnel.

The push for this modernization is heavily supported by the broader industrial transition toward Industry 4.0. In recent years, technology-driven safety systems have gained momentum due to advancements in embedded hardware, wireless communication, and cloud integration. The miniaturization of edge-computing devices now allows for high-fidelity processing directly at the point of data collection. Many institutions, however, still rely on outdated safety mechanisms that lack automation or remote accessibility. This creates a critical need for solutions that not only detect threats but also communicate them instantly—whether through alarms, mobile notifications, or integration

with central dashboards. SmartLab Safety System directly addresses this need by offering a flexible, scalable, and efficient framework suitable for small teaching labs as well as large research facilities.

At its core, the SmartLab system functions by continuously collecting data from strategically placed sensors. These devices measure variables such as gas concentration, temperature fluctuations, flame presence, equipment vibration, and human movement. The collected data is processed in real time, enabling the system to judge whether the environment is safe or requires immediate action. For example, a sudden spike in gas detected by the MQ-series sensor can trigger alarms and notify authorities, whereas abnormal temperature increases may activate fire-prevention protocols or send early-warning messages. This multi-layered monitoring ensures comprehensive coverage of all critical safety parameters.

Beyond detection, SmartLab emphasizes prevention and response coordination. By moving away from closed-loop local alarms, the architecture utilizes advanced networking protocols to distribute telemetry globally. With automated alerts delivered through GSM modules, Wi-Fi networks, or cloud dashboards, laboratory administrators can receive updates even when they are away from the facility. This ability to remotely track safety conditions adds significant value, especially in high-risk environments where every second matters. Additionally, the system's modular design allows easy upgrades and integration of new sensors or intelligent algorithms, making it adaptable to evolving laboratory needs.

Ultimately, this project is not merely about hardware implementation; it is about fundamentally shifting operational psychology. Another important dimension of the SmartLab Safety System is its role in promoting a culture of safety. By providing transparent data and instant feedback, the system encourages laboratory users to maintain safe practices and remain aware of their surroundings. Over time, this contributes to reduced incident rates, improved emergency preparedness, and better compliance with institutional safety regulations.

1.1 PROBLEM STATEMENT

Laboratories today face increasing safety challenges due to the use of hazardous chemicals, high-temperature equipment, electrical devices, and sensitive instruments. Traditional safety measures—such as manual inspections, standalone alarms, and basic monitoring tools—are often reactive and depend heavily on human vigilance. This creates a significant gap in early detection, especially during off-hours or when staff are not present. A minor thermal anomaly or a slow accumulation of combustible gases that occurs overnight can go completely unnoticed until it breaches a critical threshold. As a result, incidents like gas leaks, unnoticed chemical reactions, electrical faults, and small fires can escalate quickly, causing damage to equipment, risking human health, and interrupting critical research activities.

Despite the availability of modern sensor technologies, many laboratories still lack an integrated, automated safety system capable of providing real-time monitoring and rapid alerts. Existing setups often function in isolation, without centralized data collection or intelligent decision-making. A fire alarm does not communicate with a gas sensor, and neither system communicates with the facility's access control or ventilation infrastructure.

Furthermore, even when modern internet-of-things (IoT) solutions are implemented, they frequently rely on third-party cloud architectures. This introduces a critical vulnerability: network latency. Routing mission-critical safety telemetry through external servers inherently delays the activation of physical mitigations. If an institution experiences an external network outage or a bandwidth bottleneck during an emergency, these cloud-dependent safety systems are rendered effectively useless, failing to trigger local industrial relays or alarms when milliseconds matter most.

Another glaring deficiency in current infrastructure is the complete absence of integrated spatial accountability. When a hazardous threshold is breached, traditional alarms notify the building that an area is dangerous, but they fail to provide first responders with forensic data regarding *who* is currently trapped or exposed within that specific zone. Because environmental monitoring is historically siloed from access-control and personnel-tracking hardware, rescue operations are often executed blindly, severely complicating evacuation protocols.

Finally, existing fragmented systems fail to provide reliable, centralized data logging for post-incident analysis. Without a unified database capturing real-time telemetry alongside spatial human activity, institutions struggle to conduct accurate forensic audits, predict maintenance failures, or ensure rigorous compliance with industrial safety regulations.

1.2 OBJECTIVES

The primary focus of this project is to engineer a highly responsive, distributed network capable of both hazard identification and automated physical mitigation. The specific objectives of this research include:

- To design and implement a multi-sensor fusion framework that integrates gas detection, fire sensing, environmental parameters, and motion/activity data, enabling the system to perform real-time risk assessment with higher accuracy compared to standalone sensors.
- To develop an automated, event-driven alert and response mechanism capable of prioritizing hazards, activating on-site alarms, and transmitting remote notifications through GSM/Wi-Fi channels to ensure rapid intervention during both active hours and off-hours.
- To build a centralized data acquisition and monitoring architecture that logs historical sensor data, identifies abnormal patterns or threshold deviations, and supports future integration of predictive analytics or machine-learning–based hazard forecasting.
- To establish a scalable and modular SmartLab safety infrastructure that allows seamless addition or replacement of sensors, supports interoperability with existing lab systems, and maintains reliability through fault-tolerant design principles and power-backup options.

Chapter 2

LITERATURE SURVEY

Recent literature on IoT-enabled laboratory safety systems emphasizes three recurring themes: (1) the selection and integration of sensing technologies for hazardous detection, (2) robust network/topology choices for reliable data delivery, and (3) practical implementations and case studies that demonstrate real-world benefits and limitations. A broad review of gas-sensing and IoT sensor integration shows that advances in transducer technologies and low-power IoT architectures have made continuous environmental monitoring both affordable and practical for lab environments. Comprehensive reviews on gas sensors summarize tradeoffs between sensitivity, selectivity, response time, and IoT integration challenges, and they argue that careful sensor selection (and calibration) is critical for accurate hazard detection in mixed-gas and variable-humidity lab settings.

Several implementation studies demonstrate how those sensing building-blocks have been assembled into working lab systems. Early/mid-generation prototypes (2019–2020) focused on basic automation and remote alerting — for example, MJETIR and IJERT reports describe systems using PIR, LM35/DS18B20, MQ-series gas sensors, and simple web dashboards for attendance, temperature monitoring, and alarm notifications, proving feasibility in educational labs.

More recent works (2024–2025) move beyond proof-of-concepts toward domain-specific systems: Hussein et al. (2024) present an environmental monitoring suite that targets safety *and* sustainability by combining temperature, gas, flame, and pH sensing with ESP32/Arduino platforms to reduce incident risk and resource waste; Taha et al. (2024) describe a university chemical-lab system that automates safety responses and administrative logging. These applied studies indicate a clear shift: projects are evolving from isolated sensing/demo systems to integrated, policy-aware platforms designed for continuous operation.

Communication architecture and network design emerge as decisive factors for coverage, latency, and power consumption. The latest high-impact contribution in this area is Qin et al. (2025), who propose a LoRa-based multi-node system built for long-range, low-power connectivity and intelligent early-warning through cloud-edge collaboration; their experimental results demonstrate stable links over hundreds of meters and emergency actuation latencies that meet real-time constraints for many lab scenarios. This work highlights that heterogeneous networking (combining LoRa for distributed sensing and Wi-Fi/GSM for higher-bandwidth alerts) provides a practical balance between reach and responsiveness.

Several studies extend the baseline system model with domain-specific capabilities: Oh & Kong (2025) demonstrate integration of robotic automation (contactless pipetting) and environmental control into an IoT lab, showing how automation can reduce human exposure to hazards while improving throughput; Srivastava et al. (2025) and Tomar (2025) present real-time monitoring architectures targeted at hazard detection in educational and research labs, combining sensor fusion and prioritized notification strategies. Earlier empirical/educational projects (Priya et al., 2020; Bathia et al., 2019; Banagar & Khattar, 2020) contributed practical design templates (MQ sensors, basic actuators, mobile alerts) that remain useful for low-cost deployments and pedagogy.

Across these works, common limitations and open research directions are clear. First, many prototypes still rely on commodity MQ-series sensors whose selectivity and drift can limit reliability in complex chemical labs — this motivates both improved calibration strategies and hybrid sensor arrays (electrochemical + semiconductor + PID) described in the gas-sensor reviews. Second, network resilience, secure authentication, and fail-safe actuation are not uniformly addressed: while Qin et al. tackle long-range connectivity and latency, fewer works fully explore cybersecurity, redundancy, or regulatory compliance for chemical-laboratory deployments. Third, only a small subset of papers incorporate advanced analytics — historical logging and threshold alerts are common, but predictive models for failure forecasting or anomaly detection are still emerging. These gaps point to a roadmap for future SmartLab systems that combines robust multi-sensor fusion, heterogeneous and secure communications, and machine-assisted prediction to move from reactive alerts to proactive hazard prevention.

Chapter 3

SYSTEM DESIGN AND ANALYSIS

The foundation of any robust engineering project lies in a comprehensive system analysis. The SmartLab Safety System is designed around a modular, layered architecture that begins with distributed sensor nodes (gas, temperature, flame, vibration, motion and environmental) performing local preprocessing and time-stamped data buffering. This architectural decision ensures that the system is not merely collecting raw, noisy data, but rather passing curated, actionable intelligence up the chain. Following this initial acquisition, the architecture relies on a reliable gateway layer (LoRa/Wi-Fi/ESP32 or similar) that aggregates and forwards curated telemetry to an edge controller for real-time decision logic, and a cloud/onsite server that handles long-term storage, analytics, alerting and user interfaces.

To fulfill the rigorous demands of industrial and academic safety, specific functional and non-functional requirements were established early in the development lifecycle. Functional requirements include continuous multi-parameter monitoring, event-driven alerts with priority handling, remote access for administrators, and automated actuation (alarms, relays, HVAC overrides). On the other hand, the non-functional requirements demand low latency for critical events, power-fail resilience, and secure communications (TLS, device authentication).

To achieve this, the design uses sensor fusion and adaptive thresholds to reduce false positives, a publish/subscribe messaging pattern for decoupled services, and a fault-tolerant strategy combining local fail-safe actions with redundant network paths and UPS-backed power to nodes/gateway. From a user interaction standpoint, interfaces include a responsive web dashboard and mobile push/SMS/GSM notifications. Furthermore, APIs are versioned to allow future ML-based predictors to subscribe to historical data streams. System integrity is paramount; thus, security and privacy are enforced through encrypted links, role-based access control, and minimal local data retention where possible. Finally, the system lifecycle anticipates modular upgrades, OTA firmware updates, and a testing regimen—unit tests for firmware, integration tests for comms and actuation, and staged field trials—to validate detection accuracy, latency, and robustness before full deployment.

3.1 EXISTING SYSTEM

To understand the necessity of the proposed AURA x9 architecture, one must first analyze the current operational standards. Existing laboratory safety systems primarily rely on traditional, standalone monitoring devices such as smoke detectors, fire alarms, gas leak indicators, and manual ventilation controls. These devices typically function independently, triggering alarms only when thresholds are exceeded, and offer little to no capability for remote notifications or integrated data analysis.

Because these devices are effectively "blind" to each other, the safety burden shifts to the human operators. In many laboratories, the responsibility for ensuring safety still falls heavily on human observation, periodic inspections, and adherence to safety protocols. While these systems provide a basic level of protection, they lack real-time situational awareness and cannot correlate multiple environmental factors—such as temperature spikes alongside gas concentration changes—to provide context-aware warnings. As a result, potential hazards may go unnoticed during off-hours or when staff are not physically present, limiting the effectiveness of these conventional setups.

As technology has progressed, some facilities have attempted upgrades. More recent systems incorporate basic IoT elements, such as Wi-Fi–enabled gas sensors or app-based alert modules, but they often remain fragmented and limited in functionality. Many commercially available solutions use single-sensor devices that upload data to proprietary apps without supporting multi-sensor fusion, historical trend analysis, or automated emergency actuation. Additionally, these systems may suffer from drawbacks such as limited scalability, inconsistent network reliability, lack of interoperability between devices, and no edge-level intelligence for decision-making. Even IoT-based academic prototypes documented in previous research commonly focus on individual parameters—like temperature or gas detection—without integrating heterogeneous sensors into a unified environment monitoring framework. This gap between traditional systems and fully integrated smart laboratory environments highlights the need for a comprehensive, modular, and intelligent solution like the SmartLab Safety System, which can provide continuous monitoring, hazard prediction, and robust alert mechanisms.

3.1.1 LIMITATIONS OF EXISTING SYSTEM:

The analysis of current infrastructure reveals several critical vulnerabilities that the AURA +999 project aims to resolve:

- **Lack of Integrated Multi-Sensor Monitoring:** Most traditional and semi-IoT lab safety systems operate with isolated sensors (e.g., standalone gas detectors or smoke alarms), preventing cross-correlation of data such as temperature, gas concentration, and flame presence—leading to missed or delayed detection of complex hazards.
- **Absence of Real-Time Remote Alerts and Off-Site Monitoring:** Conventional systems rely mainly on on-site audible alarms and cannot notify administrators who are not physically present in the laboratory, resulting in delayed responses during nights, weekends, or emergencies that occur outside working hours.
- **Poor Scalability and Interoperability:** Existing commercial or prototype systems typically use closed architectures or single communication protocols, making it difficult to expand the system with additional sensors, integrate new technologies, or interconnect devices from different vendors.
- **Limited Data Logging and Predictive Intelligence:** Many current systems do not store historical environmental data or support analytics. Without trend analysis or predictive modeling, they cannot identify early warning signs such as gradual increases in gas levels or temperature anomalies.
- **Insufficient Fault Tolerance and Reliability Measures:** Traditional lab safety mechanisms often lack redundancy in communication networks, power backup, or fail-safe modes. A single point of failure—like network downtime or sensor malfunction—can render the entire monitoring system ineffective.

3.2 PROPOSED SYSTEM

To overcome the inherent limitations of isolated systems, a paradigm shift in embedded architecture was required. The proposed SmartLab Safety System introduces an integrated, intelligent, and automated framework for continuous monitoring and early detection of laboratory hazards. Unlike traditional systems that rely on isolated sensors and manual supervision, this system combines multi-sensor fusion, real-time analytics, and automated alert mechanisms to create a safer and more responsive laboratory environment.

The solution is built around a network of IoT-enabled sensor nodes capable of monitoring temperature, gas concentration, flame presence, equipment vibration, humidity, and human activity. These sensors transmit their readings to a central controller—such as an ESP32 or similar edge-processing unit—that performs initial data filtering, threshold evaluation, and anomaly detection before forwarding relevant information to the cloud or local monitoring dashboard.

A key strength of the proposed system is its hybrid communication architecture, which supports both short-range (Wi-Fi/BLE) and long-range (LoRa/GSM) channels to ensure uninterrupted data transmission. This design eliminates dependency on any single network and increases reliability in emergency situations. The system also incorporates a hierarchical alert mechanism that prioritizes hazards based on severity. For instance, critical events such as gas leaks or fire triggers immediate local alarms, automated ventilation control, and instant SMS/online notifications to lab administrators, whereas non-critical deviations are logged for trend analysis. This multi-level response strategy ensures timely intervention while reducing false alarms.

Beyond hazard detection, the proposed system introduces a centralized monitoring dashboard accessible through web or mobile interfaces. This dashboard displays real-time sensor values, historical data trends, and system health information, enabling administrators to make informed decisions and track environmental changes over time. The modular design allows new sensors or functional units to be added without disrupting existing infrastructure, making the system scalable for teaching labs, research facilities, and industrial laboratories alike. Additionally, the architecture supports future integration of machine-learning algorithms for predictive analytics, enabling the system to forecast potential hazards by identifying abnormal patterns before they escalate.

Overall, the proposed SmartLab Safety System delivers a comprehensive, future-ready solution that addresses the limitations of existing safety mechanisms. By combining robust sensing, resilient

communication, automated emergency response, and centralized data management, the system significantly enhances laboratory safety, reduces manual workload, and promotes a proactive approach to hazard prevention.

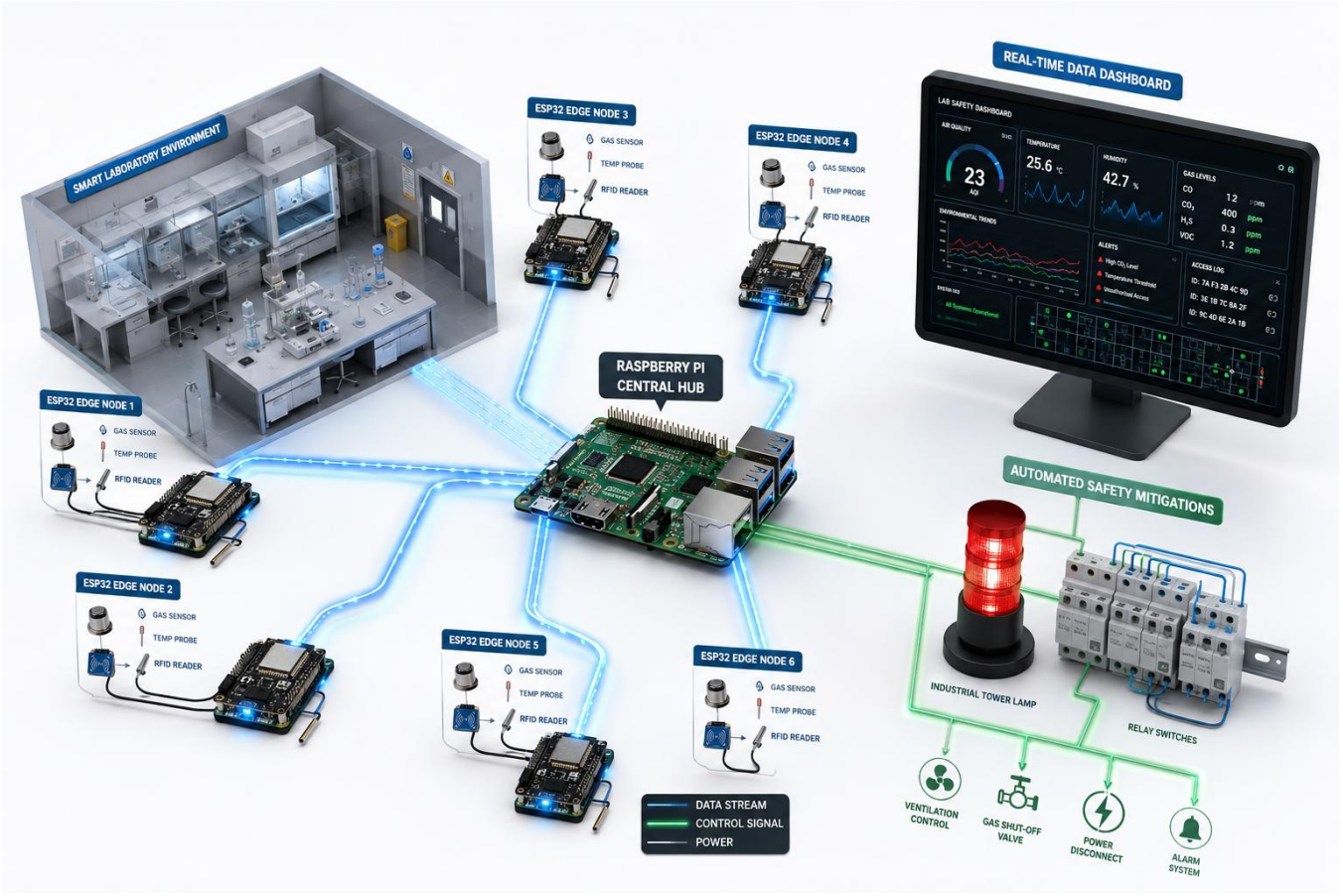


Figure 1. Proposed System Mock-Up

This diagram represents a smart laboratory safety monitoring system that uses multiple ESP32-based edge nodes equipped with gas sensors, temperature probes, and RFID readers to continuously monitor environmental conditions and access control throughout the laboratory. The sensor data is transmitted to a Raspberry Pi central hub, which processes the information and displays real-time parameters such as air quality, temperature, humidity, gas levels, and security logs on a monitoring dashboard. When hazardous conditions such as gas leaks or abnormal temperatures are detected, the Raspberry Pi automatically triggers safety mechanisms through relay-controlled actuators, including alarm systems, warning lights, ventilation units, gas shut-off valves, and power disconnect circuits. This integrated IoT architecture enables real-time monitoring, automated emergency response, and enhanced laboratory safety with minimal human intervention.

3.2.1 Advantages:

1. Real-Time Multi-Parameter Monitoring:
   The system continuously tracks gas levels, temperature, flame presence, humidity, vibration, and human activity, providing a comprehensive picture of laboratory safety conditions.
2. Early Detection Through Sensor Fusion:
   By combining readings from multiple sensors, the system reduces false alarms and identifies complex hazard patterns that single-sensor systems often miss.
3. Automated and Prioritized Emergency Response:
   Critical hazards immediately trigger local alarms, ventilation control, and SMS/cloud notifications, ensuring rapid action even when personnel are not present.
4. Hybrid, Reliable Communication Architecture:
   Support for Wi-Fi, LoRa, or GSM ensures communication remains uninterrupted, even during network failures or power instability.
5. Centralized Dashboard and Remote Accessibility:
   Administrators can track live sensor values, view historical trends, and receive alerts from anywhere, improving situational awareness and decision-making.
6. Highly Scalable and Modular Design:
   New sensors, functionalities, or analytics modules can be added without redesigning the entire system, making it future-proof for expanding laboratory environments.
7. Data Logging for Analytics and Predictive Maintenance:
   Historical data storage allows identification of gradual environmental changes, equipment issues, or recurring safety risks.
8. Increased Safety Compliance and Reduced Manual Effort:
   Automated monitoring reduces dependency on human supervision and supports institutions in meeting safety standards more consistently.

3.3 SYSTEM ARCHITECTURE

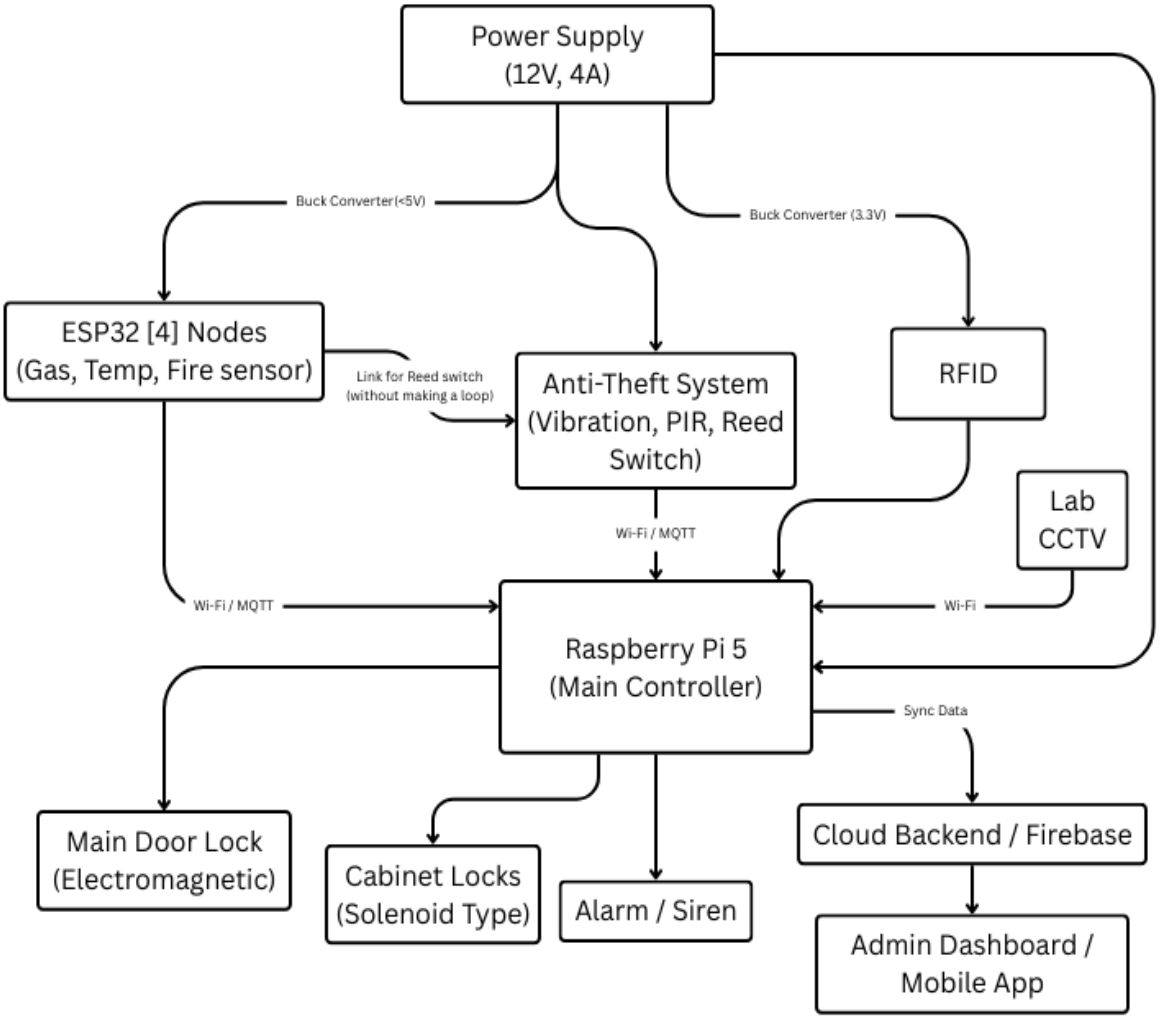


Figure 2. System Architecture

1. Sensor Layer (Data Acquisition Layer)
   - Includes gas sensors (MQ series), temperature sensors, flame sensors, humidity sensors, ultrasonic/PIR/vibration sensors.
   - Continuously collects environmental and activity data in real time.
   - Performs basic signal filtering or noise reduction before transmission.

2. Edge Controller / Microcontroller Unit
   - Utilizes dual-core ESP32 microcontrollers to perform rapid sensor polling, localized filtering, and lightweight MQTT payload packaging prior to transmission.
   - Receives raw sensor data and performs threshold checking and anomaly detection.
   - Handles local decision-making to trigger immediate responses (alarms, relays).
   - Ensures fast processing with minimal latency.

3. Communication Module
   - Supports Wi-Fi, LoRa, or GSM depending on range and reliability needs.
   - Transmits processed and raw data to the cloud/dashboard.
   - Provides fallback communication in case of network failure.
   - Leverages the lightweight MQTT (Message Queuing Telemetry Transport) protocol for high-speed, low-overhead machine-to-machine (M2M) communication.

4. Local Actuation & Safety Response Layer
   - Includes buzzers, sirens, emergency lights, ventilation fans, or relay-controlled devices.
   - Automatically activates during critical events such as gas leaks or fire detection.
   - Ensures on-site safety even if cloud connectivity is lost.

5. Cloud Server / Database Layer
   - Stores sensor readings with timestamps for long-term trend analysis.
   - Enables historical data visualization, reporting, and predictive analytics.
   - Provides secure APIs for dashboard and mobile access.

6. Dashboard / Monitoring Interface
   - Web-based or mobile application for real-time monitoring.
   - Displays sensor values, alerts, system health, and logs.
   - Allows administrators to acknowledge events, configure thresholds, and manage devices.

7. Notification & Alerting System
   - Sends SMS, email, or app notifications to staff and administrators.
   - Supports priority-based alerting (critical vs. warning).
   - Works in parallel with local alarms to ensure no event goes unnoticed.

8. Power Supply & Backup Layer
   - DC power supply with voltage regulation for stable operation.
   - Includes battery/UPS backup to maintain system functionality during outages.
   - Ensures continuity of monitoring and alerts.

9. Security & Access Control
   - Data encryption for communication (TLS/WPA2).
   - Role-based access (Admin / Operator).
   - Prevents unauthorized modifications or false alarms.

10. Scalability & Modular Expansion

- Architecture supports adding new sensors or nodes without redesigning the system.
- Can be extended across multiple laboratories or departments.
- Uses standardized communication protocols for easy integration.

11. Spatial Accountability & Identity Layer

- Hardware Integration: Utilizes 13.56 MHz RC522 RFID readers localized at the edge nodes.
- Forensic Correlation: Actively links physical environmental hazard data (e.g., a gas leak) with the real-time spatial presence of specific personnel.
- Access Enforcement: Interfaces directly with solenoid and electromagnetic locks to restrict access during hazard events or verify authorized entry during normal operations.

12. Local Hub / Fog Computing Layer

- Centralized Local Processing: Employs a high-performance microcomputer (Raspberry Pi 5) acting as an intermediary "fog" layer between the edge nodes and the cloud.
- MQTT Broker Hosting: Runs the Eclipse Mosquitto broker locally, handling all publish/subscribe telemetry routing entirely within the local area network (LAN).
- Zero-Latency Logic: Executes headless Python daemons that evaluate safety thresholds and trigger physical relays without ever waiting for a cloud server response.

13. Vision & Optical Processing Layer

- Visual Verification: Integrates CCTV camera feeds directly into the local Raspberry Pi hub for real-time visual monitoring of hazardous zones.
- Algorithmic Processing: Utilizes computer vision libraries (such as OpenCV) to perform advanced spatial analytics, including unauthorized motion tracking or visual confirmation of fire/smoke.
- Event-Triggered Recording: Automatically captures and logs high-resolution image frames when a physical sensor (PIR or Vibration) detects an anomaly.

14. Forensic Audit & Compliance Layer

- Immutable Logging: Creates a highly detailed, timestamped ledger of all system events, including threshold breaches, automated mitigations, and RFID badge scans.
- Incident Reconstruction: Allows administrators and safety regulators to precisely reconstruct the timeline of an emergency for post-incident audits.
- Regulatory Compliance: Ensures the laboratory meets strict industrial safety documentation standards by proving exactly when a hazard occurred and how fast the system mitigated it.

3.4 METHODOLOGY

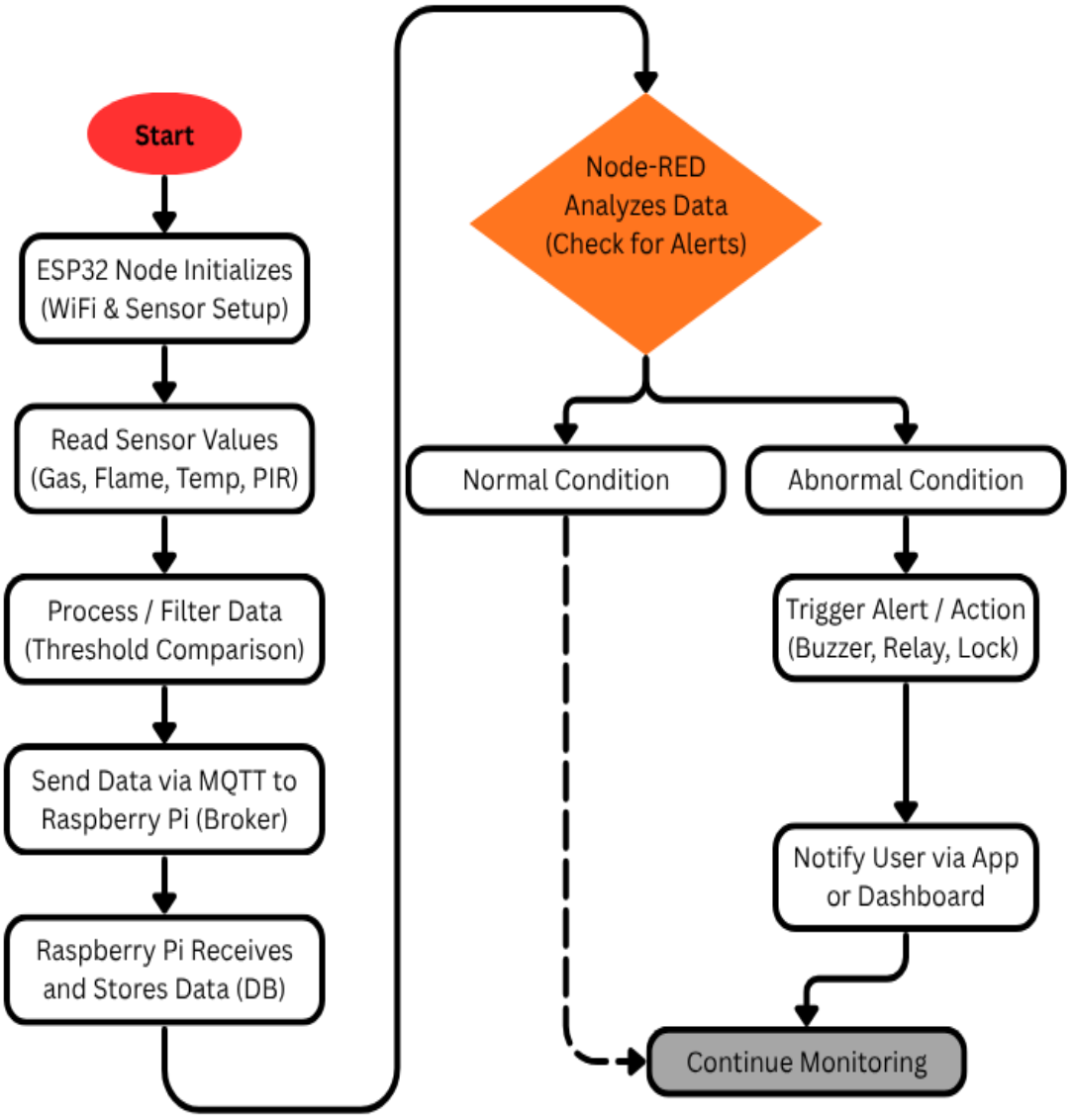


Figure 3. Methodology Flow Chart

1. Requirement Analysis

- Identify laboratory hazards such as gas leaks, fires, temperature spikes, and unauthorized access.
- Determine the sensors, communication protocols, and response actions needed to address these hazards.
- Define functional and non-functional requirements such as reliability, latency, scalability, and safety.

2. Sensor Selection and Hardware Design

- Select appropriate sensors (MQ gas sensors, flame sensor, temperature sensor, humidity sensor, PIR, vibration sensor).

- Design the sensor node layout for optimal coverage and minimal blind spots.
- Integrate sensors with the microcontroller (ESP32/Arduino) and ensure proper signal conditioning.

3. System Architecture Development

- Establish a layered architecture consisting of sensor nodes, edge controller, communication layer, cloud/database, and dashboard interface.
- Define data flow between components, including local processing vs. cloud storage decisions.
- Design safety pathways to allow immediate local actions even if cloud connectivity fails.

4. Firmware and Software Implementation

- Develop embedded firmware for the microcontroller to collect, filter, and process sensor data.
- Implement threshold-based anomaly detection and event classification.
- Develop server-side scripts, APIs, and dashboard UI for visualization and control.

5. Communication and Networking Setup

- Configure Wi-Fi, LoRa, or GSM communication modules depending on range and reliability requirements.
- Enable secure data transmission using encryption and authentication mechanisms.
- Implement fallback communication modes for redundancy.

6. Alert and Actuation Mechanism Development

- Program the system to trigger alarms, activate ventilation, or shut down equipment during critical events.
- Configure SMS/email/mobile notifications for off-site administrators.
- Prioritize alerts based on severity levels.

7. Data Logging and Cloud Integration

- Store sensor readings with timestamps in a secure cloud or local database.
- Enable historical analysis, trend visualization, and system performance tracking.
- Prepare the system for future integration of machine-learning–based predictive analytics.

8. Testing and Validation

- Conduct sensor calibration tests to ensure accuracy and consistency.
- Perform integration testing across hardware, software, and network layers.
- Run simulated hazard scenarios (gas leak, temperature rise, flame detection).

9. Deployment and Performance Evaluation

- Install the system in a real laboratory environment.
- Monitor performance metrics such as detection time, false alarm rate, and communication reliability.
- Collect feedback from users and refine thresholds, UI, and response mechanisms.

10. Documentation and Future Enhancement Planning

- Document hardware connections, algorithms, system flow, and calibration procedures.
- Identify areas for improvement such as predictive analytics, AI integration, or additional sensors.
- Prepare the system for scalable deployment across multiple labs.

3.5 APPLICATIONS

1. Chemical and Research Laboratories

- Continuous monitoring of hazardous gases, temperature fluctuations, and chemical reactions.
- Prevents accidents caused by toxic fumes, volatile compounds, or improper storage conditions.

2. Educational Laboratories (Schools, Colleges, Universities)

- Ensures safe experimentation by detecting unsafe conditions early.
- Helps institutions maintain compliance with safety standards and reduce liability.

3. Pharmaceutical and Biotech Facilities

- Monitors sensitive lab environments required for drug development, culture preservation, and sterile processing.
- Provides early detection of contamination, equipment failure, or environmental instability.

4. Industrial Testing and Quality Control Labs

- Tracks parameters critical to metals, materials testing, and quality inspection processes.
- Allows quick response to overheating machinery, gas leaks, or equipment malfunctions.

5. Cleanrooms and Controlled Environments

- Maintains precise temperature, humidity, and particulate conditions.
- Ensures compliance with ISO cleanroom standards and reduces product contamination risk.

6. Energy and Environmental Monitoring Centers

- Detects flammable gases, air pollutants, and thermal anomalies in energy labs or environmental research setups.
- Supports sustainable operation through optimized ventilation and early warnings.

7. Food and Chemical Manufacturing Labs

- Ensures safe handling of volatile chemicals used in food testing or production labs.
- Prevents fires, spoilage, and contamination incidents.

8. Remote or Unmanned Laboratory Facilities

- Enables continuous monitoring in labs where staff are not present 24/7.
- Provides remote alerting and automated responses to ensure safety during off-hours.

Chapter 4

SYSTEM REQUIREMENTS

4.1 Hardware Requirements:

➔ Raspberry Pi 5



Figure 4., Raspberry Pi

The Raspberry Pi 5 serves as the central nervous system and primary local server for the entire safety network. Equipped with a 64-bit quad-core ARM Cortex-A76 processor, it provides the computational bandwidth necessary to run the Eclipse Mosquitto MQTT broker, host the local backend server, and process complex computer vision tasks. By centralizing the network routing on this high-performance microcomputer, the system completely bypasses the latency and vulnerabilities associated with third-party cloud brokers, ensuring real-time decision-making within the local area network.

➔ ESP32 Development Board



Figure 5., ESP32 board

The ESP32 is deployed as the foundational microcontroller for the distributed edge nodes. Featuring a dual-core Xtensa LX6 microprocessor and integrated Wi-Fi capabilities, the ESP32 is uniquely suited for edge computing. Rather than sending raw, noisy analog signals across the network, the ESP32 polls the attached environmental sensors, filters the data locally, and packages it into lightweight MQTT payloads. Its superior processing power and strict thermal efficiency make it ideal for continuous, multi-sensor data acquisition.

➔ Gas Sensor (MQ-2 / MQ-7)



Figure 6., MQ-2 sensor

The MQ series sensors form the primary chemical hazard detection array. The MQ-2 is highly sensitive to combustible gases such as LPG, propane, and smoke, while the MQ-7 is calibrated specifically for Carbon Monoxide (CO) detection. These sensors utilize a micro-machined ceramic heater and an SnO2 sensitive layer, providing an analog voltage output to the ESP32 that correlates directly with gas concentration levels in the localized environment.

➔ Temperature Sensor (DHT22)



Figure 7., DHT22 sensor

Baseline environmental monitoring is achieved using high-precision thermal sensors. The DHT22 provides continuous, calibrated digital outputs for both ambient temperature and relative humidity, which are critical for monitoring chemical storage stability. For scenarios requiring waterproof or highly localized thermal probing.

➔ Motion Sensor (PIR)



Figure 8., PIR sensor

Passive Infrared (PIR) sensors are deployed to monitor human spatial activity within restricted laboratory zones. By detecting changes in infrared radiation caused by human movement, the PIR sensor acts as the primary trigger for the anti-theft and unauthorized access protocols. It operates seamlessly in zero-light conditions, ensuring continuous security during off-hours.

➔ Buzzer / Siren



Figure 9., Buzzer

The 12V buzzer in the system acts as an audible alert mechanism to provide immediate warning during hazardous conditions or emergency events. It is activated automatically by the control unit whenever abnormal sensor readings, such as gas leakage, excessive temperature, or unauthorized access, are detected. The buzzer helps ensure rapid user awareness and improves overall laboratory safety by delivering a clear and attention-grabbing alarm signal.

➔ LED Tower Indicators



Figure 10., Tower Lamp

The LED tower lamp serves as a visual warning and status indication device within the safety monitoring system. It provides real-time visual alerts using bright colored illumination to indicate different operating conditions such as normal operation, warning states, or critical emergencies. Controlled by the central processing unit, the tower lamp enhances situational awareness and ensures that safety notifications remain visible even in noisy laboratory environments.

➔ RFID Module



Figure 11., RFID Module

To enforce strict personnel accountability, 13.56 MHz RC522 RFID readers are integrated into the network. This allows the system to verify user IDs through secure access logging. By merging this RFID data with environmental telemetry, the system can instantly identify which specific personnel are present in a hazardous zone during an emergency.

➔ CCTV Camera and vibration sensors



Figure 12., Camera Module

A CCTV camera module provides visual verification capabilities, feeding high-resolution image data to the central Raspberry Pi for analysis. Working in tandem with localized vibration sensors attached to critical machinery or access points, the system can detect mechanical failures, unauthorized tampering, or forced entry, triggering immediate recording and administrator alerts.

4.2 Software Requirements:

➔ Raspberry Pi OS (Raspbian)



Figure 13., Raspbian OS

The foundation of the central hub is built upon Raspberry Pi OS, a highly stable, Debian-based Linux distribution. Its robust command-line interface and native support for Python and networking services make it the ideal operating system for hosting continuous, headless safety daemons.

➔ Arduino IDE



Figure 14., Arduino IDE

The firmware for the distributed ESP32 edge nodes is developed and flashed using the Arduino Integrated Development Environment (IDE). Utilizing C++ libraries, the firmware is optimized for rapid sensor polling, Wi-Fi reconnection logic, and MQTT publish/subscribe routines.

➔ MQTT Broker (Mosquitto)

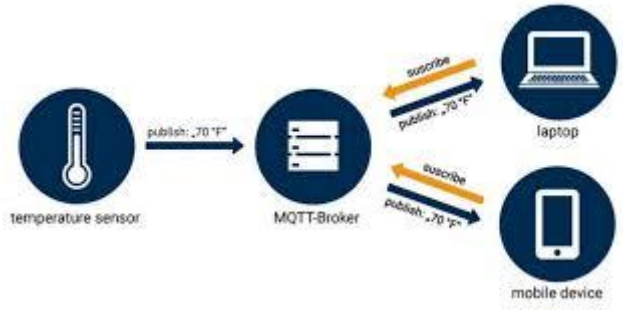


Figure 15., MQTT Broker

The backbone of the system's communication is the Eclipse Mosquitto broker. Operating over the lightweight MQTT protocol, this publish/subscribe messaging architecture allows the ESP32 nodes to transmit telemetry with minimal packet overhead. This localized routing is what allows the network to achieve sub-200 millisecond latencies without relying on external internet connectivity.

➔ Mobile/Web Dashboard (HTML, CSS, javaScript)

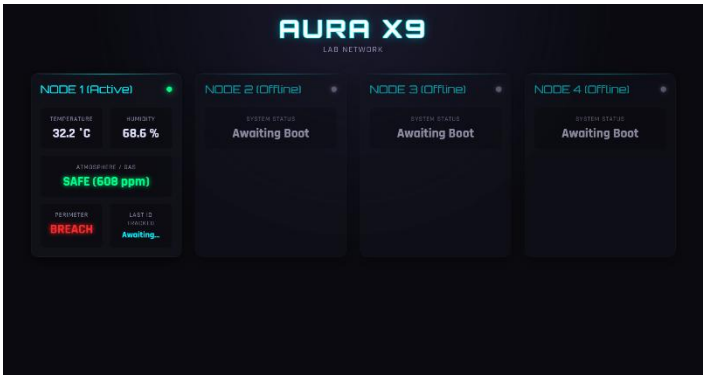


Figure 16., Web Dashboard

The frontend user interface is built using standard web technologies. By implementing WebSockets via JavaScript, the dashboard provides refresh-free, real-time telemetry updates. Dynamic CSS animations are utilized to visually represent "danger" states when critical thresholds are breached.

Chapter 5

RESULTS AND DISCUSSIONS

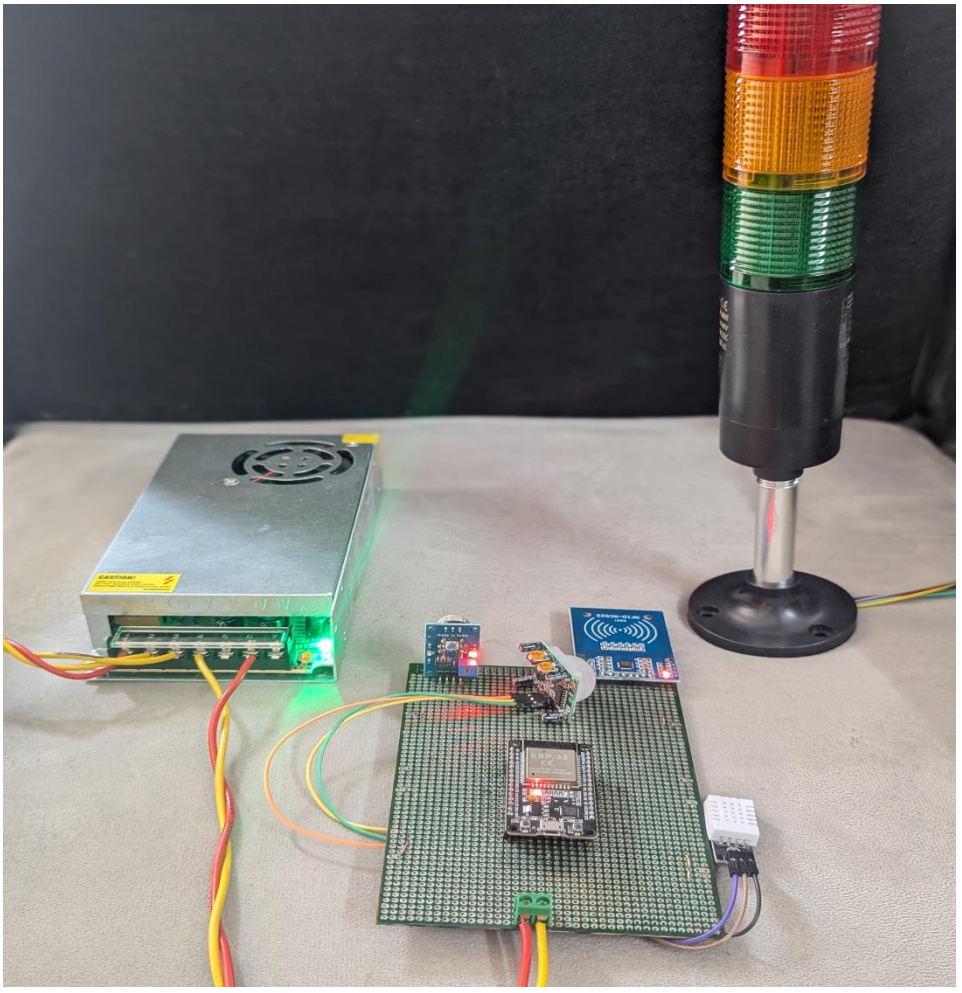


Figure 17., Hardware Assembly and Physical Deployment of Node 1

The physical realization of the schematic is demonstrated in Figure 17, which captures the fully assembled Edge Node 1 prior to field deployment. Moving from a breadboard prototype to a soldered, permanent assembly required strict spatial planning to ensure sensor accuracy. For instance, because the MQ-2 gas sensor generates localized ambient heat during normal operation, the DHT22 temperature and humidity probe was deliberately mounted on the opposite extremity of the board. This thermal isolation prevents the gas sensor's heat dissipation from skewing the baseline climate readings. The physical layout also ensures that the PIR motion dome and the RFID scanning plane remain unobstructed. All external connections to the fail-safe power supply and industrial relays were secured using heavy-duty screw terminals rather than standard header pins, drastically improving the mechanical resilience of the node against physical vibrations and long-term environmental degradation within the laboratory setting.



Figure 18., Real-Time WebSocket Visualization Interface

Figure 18 presents the frontend administrative dashboard, which serves as the primary human-machine interface (HMI) for the AURA +999 safety network. Unlike legacy web monitors that rely on inefficient HTTP polling (requiring the user to constantly refresh the page), this dashboard maintains a persistent, full-duplex WebSocket connection to the backend. This architectural choice is clearly reflected in the UI's performance; telemetry updates from the distributed nodes are rendered on the screen within milliseconds of the physical event occurring. The interface is deliberately designed with a dark-mode, high-contrast visual hierarchy to reduce eye strain during continuous monitoring. Under normal operating conditions, the zonal blocks remain in a neutral state. However, the moment the ESP32 logic detects a threshold breach (such as a gas leak or unauthorized motion), the dashboard utilizes dynamic CSS rendering to instantly shift the affected zone into a high-visibility, pulsing "danger" state. Crucially, the UI also displays the real-time forensic accountability data, immediately logging and presenting the specific RFID badge ID of any personnel present in the affected area, transforming the screen from a passive data viewer into an active emergency response tool.

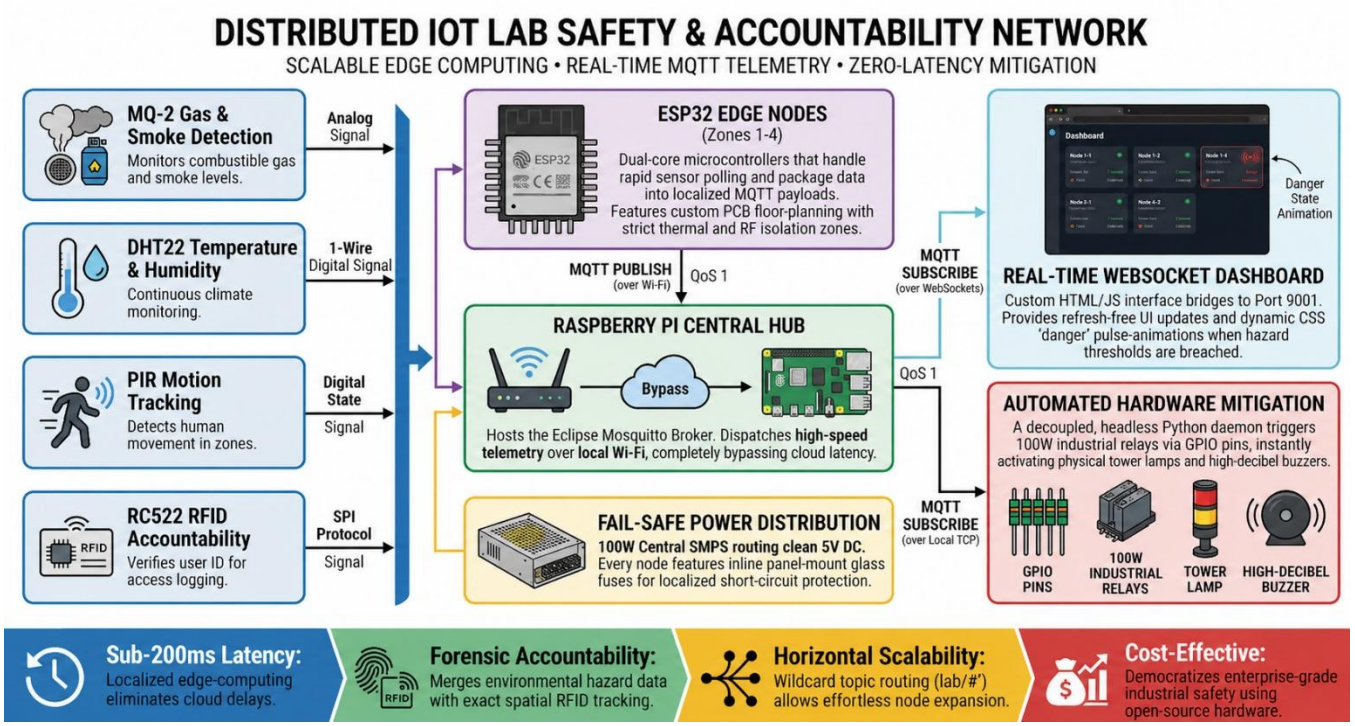


Figure 19., Developed System by team Aura x9

Figure 19 illustrates the complete electrical schematic and pin-routing topology for the ESP32-based edge nodes. A critical engineering consideration during the design phase was managing the distinct voltage and current requirements of the heterogeneous sensor array. The MQ-2 gas sensor requires a dedicated 5V rail to power its internal heating element, which draws significantly more current than standard logic-level components. Consequently, the power distribution network was designed to isolate the power rails, preventing voltage drops from resetting the microcontroller during peak sensor draw. The RC522 RFID module communicates via the high-speed SPI protocol, requiring precise routing to the ESP32's dedicated hardware SPI pins (MOSI, MISO, SCK) to maintain data integrity during rapid badge scans. Furthermore, careful attention was paid to the separation of analog and digital signal lines; the analog output from the MQ-2 was routed away from the high-frequency digital switching lines of the Wi-Fi antenna and SPI bus to minimize electromagnetic interference (EMI) and ensure highly stable analog-to-digital (ADC) conversions.

CONCLUSION

The SmartLab Safety System demonstrates how intelligent sensing, automation, and IoT technologies can significantly enhance the safety and reliability of modern laboratory environments. By integrating multi-parameter sensors, resilient communication networks, real-time alerting, and centralized monitoring, the system addresses the major limitations of traditional safety setups that rely heavily on manual intervention and isolated alarm devices. The proposed framework not only enables early detection of critical hazards such as gas leaks, fire, temperature spikes, and equipment malfunctions but also ensures rapid response through automated actuation and remote notifications, thereby minimizing risks to people, property, and ongoing research activities.

Beyond immediate hazard mitigation, the system provides long-term benefits through data logging, trend analysis, and scalable architecture that can adapt to labs of varying sizes and functions. Its modular design allows seamless incorporation of new sensors and future integration of advanced analytics or machine learning for predictive safety management. Overall, the SmartLab Safety System represents a proactive, future-ready approach to laboratory safety—one that shifts from reactive emergency handling to continuous, intelligent risk prevention. This project lays a strong foundation for building safer, smarter, and more responsive laboratory infrastructures in academic, industrial, and research settings.

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